



Micro-Project Report

Institute Code: 0078

Academic Year: 2022-23

Program: Computer Technology

Course: Mobile Application Development(MAD)

Course Code:22617 **Scheme:** I

Semester: 6

Class: TYCM-I

Date of Report: / / 2023

Title of Micro-Project: Mobile Sensor Tester.

1.0 Problem Definition: Mobile sensor tester is mobile application (tool) designed to test the functionality and accuracy of various sensors present in a mobile phone or tablet. It is used to determine whether the sensors, such as accelerometers, compasses (Orientation sensor), proximity sensors, and light sensors, are working correctly or not. The problem with mobile sensors is that they are prone to malfunction due to various reasons such as hardware failure, software bugs, or physical damage. As these sensors play a crucial role in various functions of a mobile device, including navigation, gaming, and camera operation, their proper functioning is essential.

2.0 Rationale: The development of a mobile sensor tester is to ensure that mobile devices with sensors are functioning correctly and providing accurate response. Mobile devices rely heavily on sensors for various functionalities, and any malfunction in these sensors can affect the performance and accuracy of the device.

A mobile sensor tester helps to identify any issues with the sensors present in a mobile device, such as accelerometers, Orientation sensor, proximity sensors, and light sensors. By providing a reliable and easy-to-use tool for testing the sensors, the mobile sensor tester can help to diagnose problems with the sensors quickly. Moreover, a mobile sensor tester can save time and money by allowing users to troubleshoot and resolve sensor-related issues quickly without the need for expensive repairs or replacements. It also enables manufacturers to test the functionality of sensors before releasing the mobile devices into the market, ensuring that customers receive devices that work correctly. Sensors are reliable and function correctly, providing accurate data to users.

3.0 Aim /Benefits of Micro-Project:

The Mobile Sensor Tester app aims to test the working of mobile sensors by providing a reliable and easy-to-use tool that can quickly identify any issues with the sensors and help users diagnose the problem. This can save time and money by allowing users to troubleshoot and resolve sensor-related issues quickly without the need for expensive repairs or replacements. The aim of a mobile sensor tester is to ensure that the sensors present in a mobile device are functioning correctly and accurately.

Some benefits of Mobile Sensor Tester app are:

1. Test working of sensors present in a mobile device quickly
2. Help users diagnose the problem with their mobile device's sensors
3. Allow manufacturers to test the functionality of sensors before releasing mobile devices into the market

4.0 Course Outcomes Achieved (COs):

CO601.1: Develop rich user interfaces by using layouts and controls.

CO601.2: Use User Interfaces components for android application development.

5.0 Literature Review:

A Mobile Sensor Tester App aims to test the available sensors in android mobile. Such as accelerometers, ,Light sensors, Proximity sensor , and orientation sensors. They evaluate the tool's performance and accuracy and demonstrate its effectiveness in detecting sensor issues .Its propose a portable sensor tester for mobile devices that can test various sensors such as accelerometers, gyroscopes, magnetometers, and light sensors. The tester can be used by both end-users and manufacturers to detect sensor issues and ensure accurate sensor data .A mobile sensor test platform for Android devices that can test various sensors such as accelerometers, gyroscopes, and magnetometers. They demonstrate the platform's effectiveness in detecting sensor issues and provide recommendations for improving the platform's accuracy.

6.0 Actual Methodology followed:

- a) Finalizing the project topic i.e. **Mobile Sensor Tester**.
- b) Prepared proposal of project.
- c) Submit the proposal.
- d) Distributed the work among team members.
- e) Interpret and Filter the information collected.
- f) Prepared the Project report.
- g) Submitted the report to Subject Teacher.
- h) Submitted the whole project (hard, soft copy) along with the viva.

7.0 Actual Resources used:

Sr. No.	Name of Resource/material	Specifications	Qty	Remarks
1	Laptop	Hp intel i5 Processor, 8 GB Ram. 512 GB SSD	01	Available
2	Operating System	Windows 11 x64 bit , Android	02	Available
3	Other software	Chrome, Microsoft Word, Android Studio Electric Eel 2022.1.1	01	Available
4	Android Mobile Phone	Android 9 up to latest android version	01	Available
5	Data Cable	As per mobile charging port	01	Available

8.0 Outputs of the Micro-Project:

Code:

[activity_main.xml](#)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_marginTop="120dp">

        <ImageButton
            android:id="@+id/Orientationsensor"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginLeft="60dp"
            app:srcCompat="@drawable/orientationsen"
            tools:ignore="SpeakableTextPresentCheck" />

        <ImageButton
            android:id="@+id/Proximitysensor"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginLeft="50dp"
            app:srcCompat="@drawable/proximitysen"
            tools:ignore="SpeakableTextPresentCheck" />
    </LinearLayout>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:layout_marginTop="20dp">

        <ImageButton
            android:id="@+id/Lightsensor"
```

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginLeft="60dp"
    app:srcCompat="@drawable/lightsen"
    tools:ignore="SpeakableTextPresentCheck" />
```

```
<ImageButton
    android:id="@+id/Accelerometersensor"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginLeft="50dp"
    app:srcCompat="@drawable/accsensor"
    tools:ignore="SpeakableTextPresentCheck" />
</LinearLayout>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginTop="20dp">
```

```
<ImageButton
    android:id="@+id/Camera"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginLeft="60dp"
    app:srcCompat="@drawable/cam"
    tools:ignore="SpeakableTextPresentCheck" />
```

```
<ImageButton
    android:id="@+id/Flashlight"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginLeft="50dp"
    app:srcCompat="@drawable/flashlight"
    tools:ignore="SpeakableTextPresentCheck" />
</LinearLayout>
```

```
</LinearLayout>
```

[MainActivity.java](#)

```
package com.example.mobilesensors;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.content.Intent;
```

```

import android.os.Bundle;
import android.view.View;
import android.widget.ImageButton;

public class MainActivity extends AppCompatActivity {
    ImageButton Os, Ps, Ls, Ass, Cam, flash;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Os=findViewById(R.id.Orientationsensor);
        Ps=findViewById(R.id.Proximitysensor);
        Ls=findViewById(R.id.Lightsensor);
        Ass=findViewById(R.id.Accelerometersensor);
        Cam=findViewById(R.id.Camera);
        flash=findViewById(R.id.Flashlight);

        Os.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent in1 = new Intent(getApplicationContext(),Ori_sen.class);
                startActivity(in1);
            }
        });

        Ps.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent in2 = new Intent(getApplicationContext(),Prox_sen.class);
                startActivity(in2);
            }
        });

        Ls.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent in3 = new Intent(getApplicationContext(),Light_sen.class);
                startActivity(in3);
            }
        });

        Ass.setOnClickListener(new View.OnClickListener() {
            @Override

```

```

        public void onClick(View view) {
            Intent in4 = new Intent(getApplicationContext(),Acce_sen.class);
            startActivity(in4);

        }
    });
    Cam.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            Intent in5 = new Intent(MainActivity.this,Camera.class);
            startActivity(in5);

        }
    });
    flash.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            Intent in6 = null;
            if (android.os.Build.VERSION.SDK_INT >=
android.os.Build.VERSION_CODES.LOLLIPOP) {
                in6 = new Intent(getApplicationContext(), Flashlight.class);
            }
            startActivity(in6);

        }
    });
}
}

```

[orientationsen.xml](#)

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

<TextView
    android:id="@+id/tvHeading"
    android:layout_width="266dp"
    android:layout_height="56dp"
    android:text="Heading: 0.0"
    android:textColor="#E65151"
    android:textSize="25dp"

```

```
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.634"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.105" />
```

```
<ImageView
    android:id="@+id/imageViewCompass"
    android:layout_width="306dp"
    android:layout_height="279dp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.495"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/tvHeading"
    app:layout_constraintVertical_bias="0.331"
    app:srcCompat="@drawable/orisensor" />
```

```
<Button
    android:id="@+id/gb"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Go Back"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/imageViewCompass"
    app:layout_constraintVertical_bias="0.691" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

[Ori sen.java](#)

```
package com.example.mobilesensors;
import android.annotation.SuppressLint;
import android.app.Activity;
import android.content.Intent;
import android.hardware.Sensor;
import android.hardware.SensorEvent;
import android.hardware.SensorEventListener;
import android.hardware.SensorManager;
import android.os.Bundle;
import android.view.View;
import android.view.animation.Animation;
import android.view.animation.RotateAnimation;
import android.widget.Button;
import android.widget.ImageView;
```

```
import android.widget.TextView;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
public class Ori_sen extends AppCompatActivity implements SensorEventListener {  
    private ImageView image;  
    private float currentDegree = 0f;  
    private SensorManager mSensorManager;  
    TextView tvHeading;  
    Button gb;  
    @SuppressWarnings("MissingInflatedId")  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.orientationsen);  
        image = (ImageView) findViewById(R.id.imageViewCompass);  
        tvHeading = (TextView) findViewById(R.id.tvHeading);  
        mSensorManager = (SensorManager) getSystemService(SENSOR_SERVICE);  
        gb=findViewById(R.id.gb);  
        gb.setOnClickListener(new View.OnClickListener() {  
            @Override  
            public void onClick(View view) {  
                Intent intent = new Intent(Ori_sen.this,MainActivity.class);  
                startActivity(intent);  
            }  
        });  
    }  
    @Override  
    protected void onResume() {  
        super.onResume();  
        mSensorManager.registerListener(this,  
mSensorManager.getDefaultSensor(Sensor.TYPE_ORIENTATION),  
        SensorManager.SENSOR_DELAY_GAME);  
    }  
    @Override  
    protected void onPause() {  
        super.onPause();  
        mSensorManager.unregisterListener(this);  
    }  
    @Override  
    public void onSensorChanged(SensorEvent event) {  
        float degree = Math.round(event.values[0]);  
        tvHeading.setText("Heading: " + Float.toString(degree) + " degrees");  
        RotateAnimation ra = new RotateAnimation(currentDegree, -  
degree,Animation.RELATIVE_TO_SELF, 0.5f, Animation.RELATIVE_TO_SELF, 0.5f);  
        ra.setDuration(210);  
    }  
}
```



```

        ra.setFillAfter(true);
        image.startAnimation(ra);
        currentDegree = -degree;
    }
    @Override
    public void onAccuracyChanged(Sensor sensor, int accuracy) {
    }
}

```

[proximity.xml](#)

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/container"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/sensorStatusTV"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
        android:fontFamily="@font/bungee_shade"
        android:gravity="center_horizontal"
        android:textAlignment="center"
        android:textColor="#E31814"
        android:textSize="50dp" />

    <Button
        android:id="@+id/gb"
        android:layout_width="160dp"
        android:layout_height="46dp"
        android:layout_below="@+id/sensorStatusTV"
        android:layout_alignParentStart="true"
        android:layout_alignParentEnd="true"
        android:layout_alignParentBottom="true"
        android:layout_marginStart="117dp"
        android:layout_marginTop="162dp"
        android:layout_marginEnd="134dp"
        android:layout_marginBottom="92dp"
        android:text="Go Back" />

</RelativeLayout>

```

Prox_sen.java

```
package com.example.mobilesensors;
import android.annotation.SuppressLint;
import android.app.ActivityManager;
import android.content.Context;
import android.content.Intent;
import android.graphics.Color;
import android.hardware.Sensor;
import android.hardware.SensorEvent;
import android.hardware.SensorEventListener;
import android.hardware.SensorManager;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import android.view.View;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class Prox_sen extends AppCompatActivity {

    TextView sensorStatusTV;
    SensorManager sensorManager;
    Sensor proximitySensor;
    private View view;
    Button gb;

    @SuppressWarnings("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.proximity);
        sensorStatusTV = findViewById(R.id.sensorStatusTV);
        view = findViewById(R.id.sensorStatusTV);
        gb=findViewById(R.id.gb);
        gb.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent = new Intent(Prox_sen.this,MainActivity.class);
                startActivity(intent);
            }
        });
        sensorManager = (SensorManager) getSystemService(Context.SENSOR_SERVICE);
        proximitySensor = sensorManager.getDefaultSensor(Sensor.TYPE_PROXIMITY);
        if (proximitySensor == null) {
            Toast.makeText(this, "No proximity sensor found in device.", Toast.LENGTH_SHORT).show();
            finish();
        }
    }
}
```

```

    } else {
        sensorManager.registerListener(proximitySensorEventListener,
            proximitySensor,
            SensorManager.SENSOR_DELAY_NORMAL);
    }
}

private ActivityManager.TaskDescription.Builder root;
SensorEventListener proximitySensorEventListener = new SensorEventListener() {
    @Override
    public void onAccuracyChanged(Sensor sensor, int accuracy) {

    }

    @Override
    public void onSensorChanged(SensorEvent event) {
        if (event.sensor.getType() == Sensor.TYPE_PROXIMITY) {
            if (event.values[0] == 0) {
                sensorStatusTV.setText("Near ⚠ ");
                view.setBackgroundColor(Color.GREEN);
            } else {
                sensorStatusTV.setText("Away ⚠ ");
                view.setBackgroundColor(Color.YELLOW);
            }
        }
    }
};
}

```

[lightsen.xml](#)

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/tv"
        android:layout_width="206dp"
        android:layout_height="81dp"
        android:textColor="#E4130E"
        android:textSize="25dp"
        android:textStyle="bold"
        app:layout_constraintBottom_toBottomOf="parent"

```

```

app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.643"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintVertical_bias="0.48" />

```

```

<Button
    android:id="@+id/gb"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Go Back"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/tv"
    app:layout_constraintVertical_bias="0.773" />

```

```

</androidx.constraintlayout.widget.ConstraintLayout>

```

[Light sen.java](#)

```

package com.example.mobilesensors;
import androidx.appcompat.app.AppCompatActivity;
import android.annotation.SuppressLint;
import android.app.Service;
import android.content.Intent;
import android.os.Bundle;
import android.hardware.*;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

public class Light_sen extends AppCompatActivity implements SensorEventListener {
    TextView tv;
    SensorManager sensorManager;
    Sensor sensor;
    Button gb;

    @SuppressWarnings("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.lightсен);
        tv=findViewById(R.id.tv);
        gb=findViewById(R.id.gb);
        gb.setOnClickListener(new View.OnClickListener() {

```

```

        @Override
        public void onClick(View view) {
            Intent intent = new Intent(Light_sen.this, MainActivity.class);
            startActivity(intent);
        }
    });

    sensorManager =(SensorManager) getSystemService(Service.SENSOR_SERVICE);
    sensor=sensorManager.getDefaultSensor(Sensor.TYPE_LIGHT);

}
@Override
protected void onPause() {
    super.onPause();
    sensorManager.unregisterListener(this);
}
@Override
protected void onResume() {
    super.onResume();

    sensorManager.registerListener(this, sensor, SensorManager.SENSOR_DELAY_NORMAL);
}

@Override
public void onSensorChanged(SensorEvent event ) {
    if(event.sensor.getType()==Sensor.TYPE_LIGHT){
        tv.setText(" "+event.values[0]);
    }

}

@Override
public void onAccuracyChanged(Sensor sensor, int accuracy) {

}

}

```

[accelerometer.xml](#)

```

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"

```

```
tools:context=".MainActivity" >
```

```
<TextView  
    android:id="@+id/textView"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:text=""  
    tools:layout_editor_absoluteX="0dp"  
    tools:layout_editor_absoluteY="0dp" />
```

```
<Button  
    android:id="@+id/gb"  
    android:layout_width="150dp"  
    android:layout_height="54dp"  
    android:layout_alignParentStart="true"  
    android:layout_alignParentTop="true"  
    android:layout_alignParentEnd="true"  
    android:layout_alignParentBottom="true"  
    android:text="Go Back"  
    app:layout_constraintBottom_toBottomOf="@+id/textView"  
    app:layout_constraintEnd_toEndOf="@+id/textView"  
    app:layout_constraintHorizontal_bias="0.54"  
    app:layout_constraintStart_toStartOf="@+id/textView"  
    app:layout_constraintTop_toTopOf="parent"  
    app:layout_constraintVertical_bias="0.901" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

[Acce sen.java](#)

```
package com.example.mobilesensors;  
import android.annotation.SuppressLint;  
import android.app.Activity;  
import android.content.Intent;  
import android.graphics.Color;  
import android.hardware.Sensor;  
import android.hardware.SensorEvent;  
import android.hardware.SensorEventListener;  
import android.hardware.SensorManager;  
import android.os.Bundle;  
import android.view.View;  
import android.widget.Button;  
import android.widget.Toast;  
  
import androidx.appcompat.app.AppCompatActivity;  
  
public class Acce_sen extends AppCompatActivity implements SensorEventListener {  
    private SensorManager sensorManager;
```

```

private boolean isColor = false;
private View view;
private long lastUpdate;
Button gb;

@SuppressLint("MissingInflatedId")
@Override
public void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.accelerometer);

    view = findViewById(R.id.textView);
    view.setBackgroundColor(Color.YELLOW);
    gb=findViewById(R.id.gb);
    gb.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            Intent intent = new Intent(Acce_sen.this,MainActivity.class);
            startActivity(intent);
        }
    });

    sensorManager = (SensorManager) getSystemService(SENSOR_SERVICE);
    lastUpdate = System.currentTimeMillis();
}
@Override
public void onAccuracyChanged(Sensor sensor, int accuracy) {}
@Override
public void onSensorChanged(SensorEvent event) {
    if (event.sensor.getType() == Sensor.TYPE_ACCELEROMETER) {
        getAccelerometer(event);
    }
}

private void getAccelerometer(SensorEvent event) {
    float[] values = event.values;
    // Movement
    float x = values[0];
    float y = values[1];
    float z = values[2];

    float accelationSquareRoot = (x * x + y * y + z * z)
        / (SensorManager.GRAVITY_EARTH * SensorManager.GRAVITY_EARTH);

    long actualTime = System.currentTimeMillis();
    Toast.makeText(getApplicationContext(),String.valueOf(accelationSquareRoot)+" "+

```

```

        SensorManager.GRAVITY_EARTH,Toast.LENGTH_SHORT).show();

if (accelationSquareRoot >= 2) //it will be executed if you shuffle
{

    if (actualTime - lastUpdate < 200) {
        return;
    }
    lastUpdate = actualTime;//updating lastUpdate for next shuffle
    if (isColor) {
        view.setBackgroundColor(Color.YELLOW);

    } else {
        view.setBackgroundColor(Color.CYAN);
    }
    isColor = !isColor;
}
}

@Override
protected void onResume() {
    super.onResume();

    sensorManager.registerListener(this,sensorManager.getDefaultSensor(Sensor.TYPE_ACCELEROMETER
),
        SensorManager.SENSOR_DELAY_NORMAL);
}

@Override
protected void onPause() {

    super.onPause();
    sensorManager.unregisterListener(this);
}
}

```

[cam.xml](#)

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

```



```

<ImageView
    android:id="@+id/imageView"
    android:layout_width="263dp"
    android:layout_height="303dp"
    app:layout_constraintBottom_toTopOf="@+id/btn"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.681"
    app:srcCompat="@android:drawable/alert_light_frame" />

```

```

<Button
    android:id="@+id/btn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginBottom="184dp"
    android:backgroundTint="#EBE4A40C"
    android:text="Click Pic"
    android:textSize="20sp"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent" />

```

```

<Button
    android:id="@+id/gb"
    android:layout_width="139dp"
    android:layout_height="51dp"
    android:layout_marginBottom="48dp"
    android:text="Go Back"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.503"
    app:layout_constraintStart_toStartOf="parent" />

```

```

</androidx.constraintlayout.widget.ConstraintLayout>

```

[camera.java](#)

```

package com.example.mobilesensors;
import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;

```

```

import android.annotation.SuppressLint;
import android.widget.*;
import android.os.Bundle;
import android.view.View;

```

```

import android.provider.MediaStore;
import android.content.Intent;
import android.graphics.Bitmap;

public class Camera extends AppCompatActivity {
    ImageView iv;
    Button b, gb;

    @SuppressWarnings("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.cam);
        iv=findViewById(R.id.imageView);
        b=findViewById(R.id.btn);
        gb=findViewById(R.id.gb);
        gb.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent = new Intent(Camera.this,MainActivity.class);
                startActivity(intent);
            }
        });

        b.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent imagecapture =new Intent(MediaStore.ACTION_IMAGE_CAPTURE);
                startActivityForResult(imagecapture,0);

            }
        });
    }
    @Override
    protected void onActivityResult(int requestCode,int resultCode,@Nullable Intent data){
        super.onActivityResult(requestCode, resultCode, data);

        Bitmap photo=(Bitmap)data.getExtras().get("data");
        iv.setImageBitmap(photo);
    }
}

```

[flash.xml](#)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity"
    tools:ignore="HardcodedText">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:layout_marginTop="100dp"
        android:text="Flashlight"
        android:textSize="50sp"
        android:textStyle="bold|italic" />

    <ToggleButton
        android:id="@+id/toggle_flashlight"
        android:layout_width="200dp"
        android:layout_height="75dp"
        android:layout_gravity="center"
        android:layout_marginTop="32dp"
        android:onClick="toggleFlashLight"
        android:textSize="25sp" />

    <Button
        android:id="@+id/gb"
        android:layout_marginTop="350dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Go Back" />

</LinearLayout>
```

[Flashlight.java](#)

```
package com.example.mobilesensors;
import android.annotation.SuppressLint;
import android.content.Context;
import android.content.Intent;
import android.hardware.camera2.CameraAccessException;
```

```

import android.hardware.camera2.CameraManager;
import android.os.Build;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;
import android.widget.ToggleButton;
import androidx.annotation.RequiresApi;
import androidx.appcompat.app.AppCompatActivity;

@RequiresApi(api = Build.VERSION_CODES.LOLLIPOP)
public class Flashlight extends AppCompatActivity{

    private ToggleButton toggleFlashLightOnOff;
    private CameraManager cameraManager;
    private String getCameraID;
    Button gb;

    @SuppressWarnings("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.flash);
        gb=findViewById(R.id.gb);
        gb.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent = new Intent(getApplicationContext(),MainActivity.class);
                startActivity(intent);
            }
        });

        toggleFlashLightOnOff = findViewById(R.id.toggle_flashlight);

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.LOLLIPOP) {
            cameraManager = (CameraManager) getSystemService(Context.CAMERA_SERVICE);
        }

        try {
            if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.LOLLIPOP) {
                getCameraID = cameraManager.getCameraIdList()[0];
            }
        } catch (CameraAccessException e) {
            e.printStackTrace();
        }
    }
}

```

```

    }
}
@RequiresApi(api = Build.VERSION_CODES.M)
public void toggleFlashLight(View view) {
    if (toggleFlashLightOnOff.isChecked()) {

        try {

            cameraManager.setTorchMode(getCameraID, true);

            Toast.makeText(Flashlight.this, "Flashlight is turned ON", Toast.LENGTH_SHORT).show();
        } catch (CameraAccessException e) {

            e.printStackTrace();
        }
    } else {

        try {

            cameraManager.setTorchMode(getCameraID, false);

            Toast.makeText(Flashlight.this, "Flashlight is turned OFF",
Toast.LENGTH_SHORT).show();
        } catch (CameraAccessException e) {
            e.printStackTrace();
        }
    }
}

@RequiresApi(api = Build.VERSION_CODES.M)
@Override
public void finish() {
    super.finish();
    try {
        cameraManager.setTorchMode(getCameraID, false);

        Toast.makeText(Flashlight.this, "Flashlight is turned OFF",
Toast.LENGTH_SHORT).show();
    } catch (CameraAccessException e) {
        e.printStackTrace();
    }
}
}

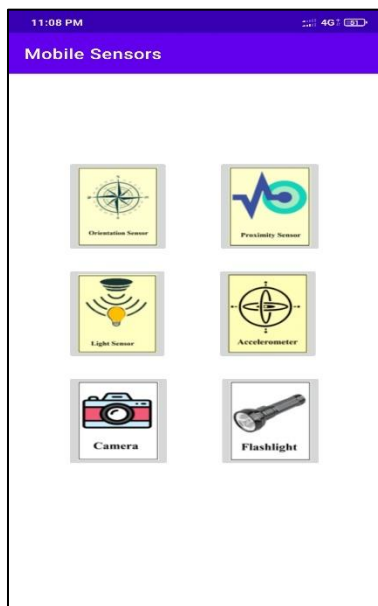
```

Result:

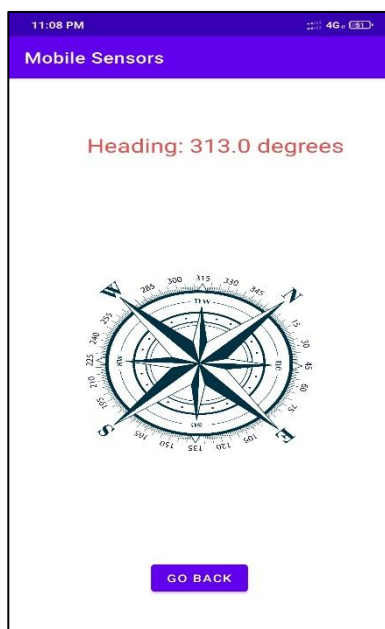
App Logo:



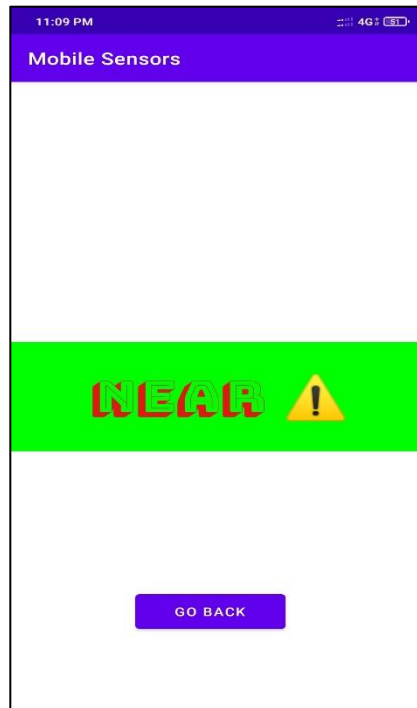
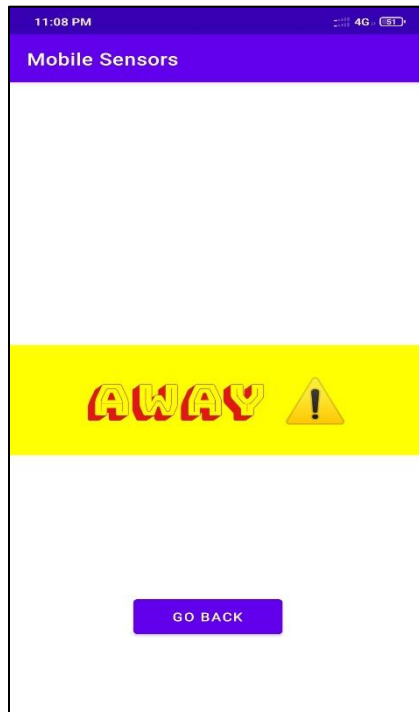
Main Screen:



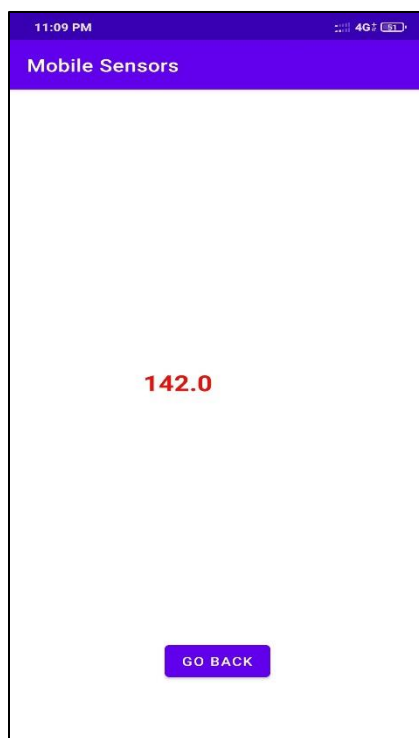
Orientation Sensor:



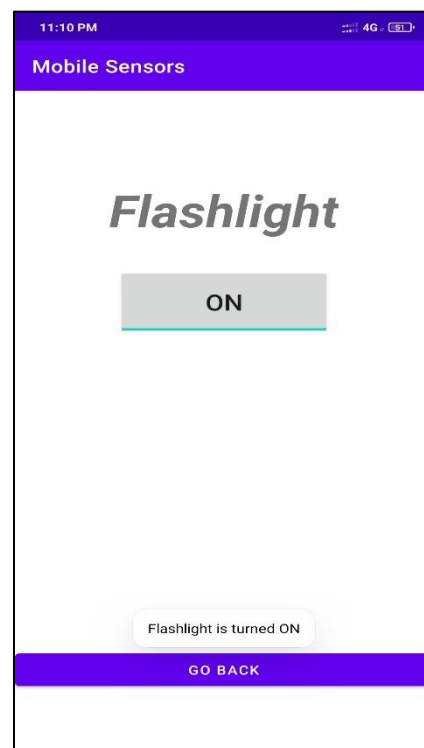
Proximity Sensor:



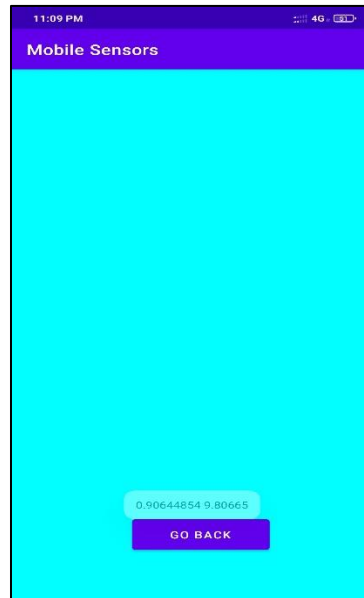
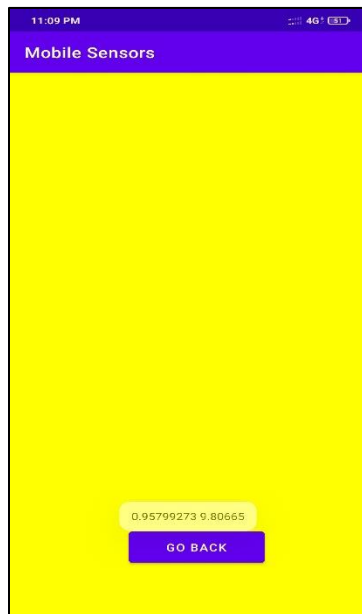
Light Sensor:



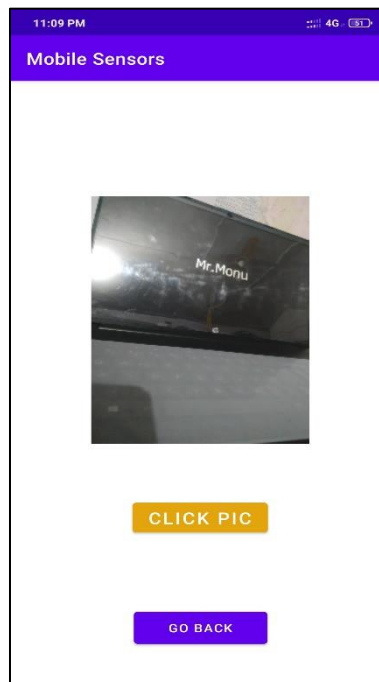
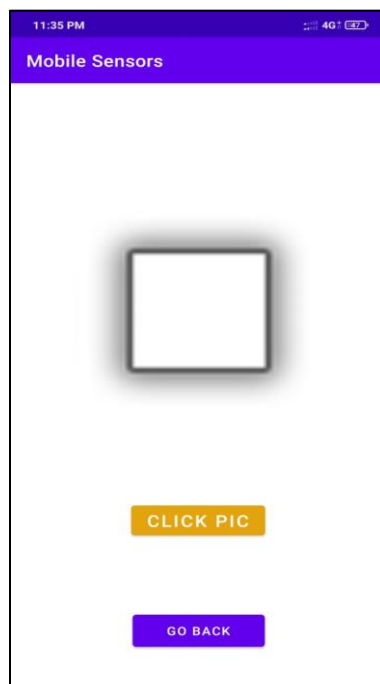
Flash Light:



Accelerometer:



Camera:



9.0 Skill Developed / Learning outcome of this Micro-Project:

Soft Skills:

- Learning basics of Android Application Development
- Problem Solving
- Leadership
- Teamwork
- Time Management
- Decision Making

Technical Skills:

- Software Efficiency
- Interacting with different sensor which is available in mobile phone.
- Word Processing
- Knowledge of performing Technical Tasks

10.0 Applications of Micro Project:

- 1) To study of Mobile Sensor Tester.
- 2) To gain basic information about Mobile Sensor Tester.
- 3) It is used to manage Mobile Sensor Tester.

11.0 Name of Group Members:

Enrolment	Roll No.	Seat No.	Name of Students	Student Signature
2100780748	69	386408	Mohan Suresh Ugale	
2100780745	67	386406	Amit Keshav More	
2000780249	52	386271	Shubham Harendra Ram	
2000780240	44	386263	Anup Jitendra Pardeshi	

Date: / 04/2023

Evaluated by: Dated Signature of Guide: _____

Name of Guide: Mrs. M. A. Shaikh